Report — Proportional Control for Maneuvering Flapping Flight

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June 17, 2026

1 Summary

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2 Model Description

2.1 Summary

We model a dragonfly-like insect with two pairs of wings. The sagittal plane is aligned with the fixed frame (x, z) plane, where z points opposite the direction of gravity. The wings in each pair beat symmetrically about the sagittal plane, and there is zero initial velocity or wind component normal to the sagittal plane, so the motion is constrained to the (x, z) plane, rendering the problem two-dimensional. A subsequent strong simplification is that the body is reduced to a point mass, so there is no coupling between body pitch and wing kinematics. In this manner we can focus entirely on the translational dynamics without concern for flight stability. The dynamical model is nondimensionalized using the body length L, the body mass m, and the gravitational time scale $\sqrt{L/g}$. The resulting dimensionless parameters, along with the wing kinematics angular parameters, are listed in Table 1. Dimensionless quantities are denoted by an asterisk $(*)$.

2.2 Aerodynamic Force

2.2.1 Model and Derivation

Each wing is modeled as a single, massless plate element of area A_w with its aerodynamic center at the $\frac{2}{3}$−span station, $r = \frac{2}{3}L_w$, whose lift and drag coefficients are prescribed by functions of the instantaneous angle of attack α. These simplified functions, given by Equation 5 and Equation 6, nevertheless capture the essential phenomena. Parameters for the wing kinematics are defined in Figure 1 and Figure 2 for a single

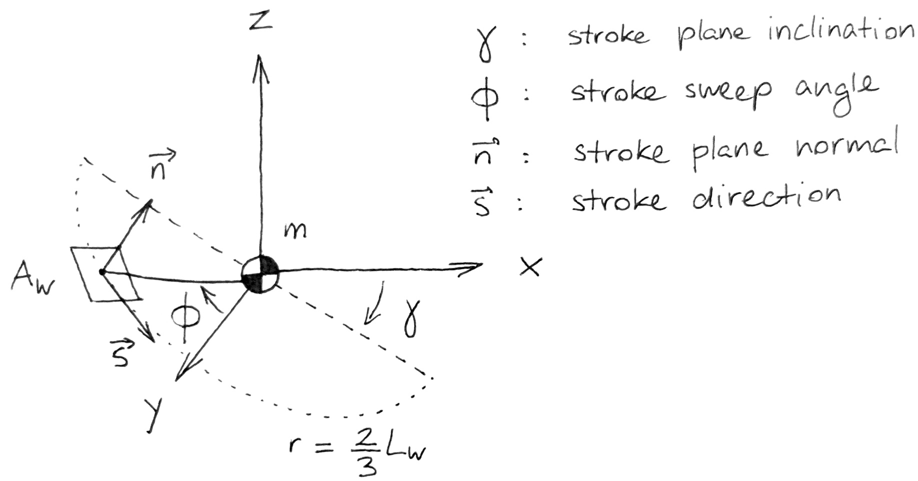

Figure 1: Wing stroke geometry

right-hand-side wing. Since the wings of each pair are assumed to beat symmetrically about the sagittal plane, the left-hand-side wing kinematics can be obtained by mirroring about the (x, z) plane.

Each wing element sweeps along an arc of radius $r = \frac{2}{3}L_w$ that lies inside the stroke plane, with normal vector $\vec{n} = \cos\gamma\vec{z} - \sin\gamma\vec{x}$, where γ is the stroke plane inclination angle relative to the vertical. The sweep angle ϕ about $\vec{n}$ is defined relative to the side vector $\vec{y}$, positive counterclockwise about $\vec{n}$. At any point along the wing stroke, the wing velocity relative to the body is

$$\vec{v}_{w/b} = \frac{2}{3}L_w\dot{\phi}\vec{s} \tag{1}$$

with $\vec{s} = \cos\phi\,(\cos\gamma\vec{x} + \sin\gamma\vec{z}) - \sin\phi\vec{y}$. The sweep angle ϕ for wing pair i is prescribed by

$$\phi^{(i)}(t) = \phi_1 \sin\left(\omega t - \sigma_0^{(i)}\right) \tag{2}$$

The phase parameter $\sigma_0^{(i)}$ is set to 0 for the forewing pair, and to σ_0 for the hindwing pair. Both wing pairs beat at the same frequency ω and with the same amplitude ϕ_1. In addition to the sweeping motion, the wing rotates about its own spanwise axis. We refer to this as the wing pitch. It is most easily described in the $(\vec{s}, \vec{n})$ plane. Figure 2 shows the pitch angle ψ defined relative to the stroke plane normal $\vec{n}$. ψ is prescribed for wing pair i by

$$\psi^{(i)}(t) = \psi_0 + \psi_1 \sin\left(\omega t - \delta_0 - \sigma_0^{(i)}\right) \tag{3}$$

Unlike for the sweep angle ϕ, the mean pitch angle ψ_0 may be nonzero, and can be used to redirect the mean aerodynamic force vector, as we will see.

It is convenient to define the chord direction vector $\vec{c} = \cos\psi\vec{n} - \sin\psi\vec{s}$. The angle of attack α is the angle between $\vec{c}$ and $\vec{v}_w$, the projection in the $(\vec{s}, \vec{n})$ plane of the wing velocity relative to the air. Care has to be taken as to the sign of α, as evidenced by the two panels in Figure 2. When $\vec{v}_w \cdot \vec{s} > \vec{c} \cdot \vec{s}$, α has the correct sign when defined from $\vec{v}_w$ to $\vec{c}$. When $\vec{v}_w \cdot \vec{s} < \vec{c} \cdot \vec{s}$, it has the correct sign when defined from $\vec{c}$ to $\vec{v}_w$. The following expression gives the correct sign:

$$\alpha = \sin^{-1}\left[\left(\frac{\vec{v}_w}{v_w} \times \vec{c}\right) \cdot (\vec{s} \times \vec{n})\right] \tag{4}$$

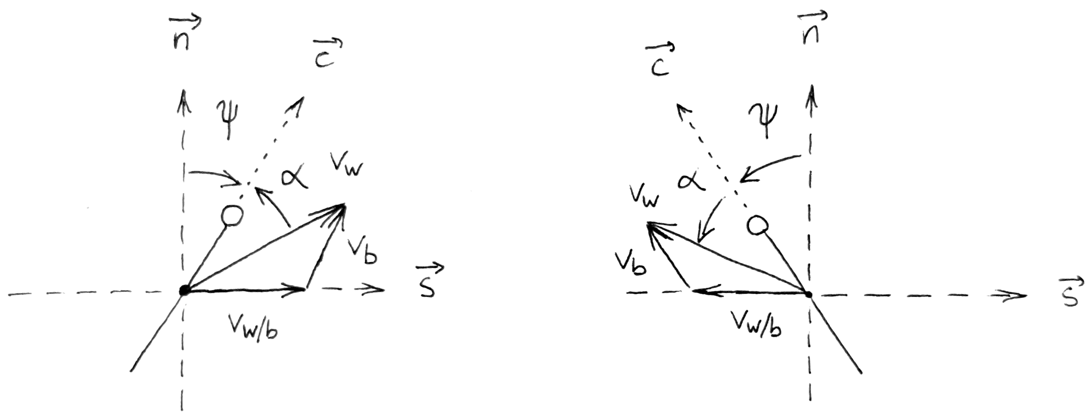

Figure 2: Wing pitch geometry and angle of attack for (a) $\vec{v}_w \cdot \vec{s} > \vec{c} \cdot \vec{s}$ and (b) $\vec{v}_w \cdot \vec{s} < \vec{c} \cdot \vec{s}$

Lift and drag coefficients are then obtained by

$$C_L(\alpha) = C_{L,0} \sin 2\alpha \tag{5}$$

$$C_D(\alpha) = C_{D,0} \cos^2 \alpha + C_{D,\frac{\pi}{2}} \sin^2 \alpha \tag{6}$$

with $C_{L,0} = 1.5$, $C_{D,0} = 0.1$, $C_{D,\frac{\pi}{2}} = 2$, as suggested by numerical simulations and experiments across the full range of angle of attack [1, 2]. The lift and drag vectors are, respectively,

$$\vec{d} = -\frac{\vec{v}_w}{v_w} \tag{7}$$

$$\vec{\ell} = \vec{d} \times (\vec{s} \times \vec{n}) \tag{8}$$

And the aerodynamic force of a single wing is

$$\vec{F}_{a,w} = \frac{1}{2}\rho_{\text{air}} A_w v_w^2 \left(C_D \vec{d} + C_L \vec{\ell} \right) \tag{9}$$

2.3 Nondimensionalization

The flying insect is subjected to gravity and the aerodynamic force on its wings. Body drag is neglected. The resulting equation of motion is

$$m\left(\ddot{x}\vec{x} + \ddot{z}\vec{z}\right) = \frac{1}{2}\rho_{\text{air}} A_w \sum_{i}^{N_w} \left[v_w^2 \left(C_D \vec{d} + C_L \vec{\ell} \right) \right]^{(i)} - mg\vec{z} \tag{10}$$

We nondimensionalize it using the body length L, the body mass m, and the gravitational time scale $\sqrt{L/g}$. The resulting dimensionless parameters, along with the already dimensionless wing angles, are summarized in Table 1. Ranges for the parameters L_w^*, A^*/m^*, and ω^* were obtained from the morphological data in [3] and the wingbeat frequency data in [4] for *Anisoptera*. The resulting dimensionless equation of motion is

$$\ddot{x}^*\vec{x} + \ddot{z}^*\vec{z} = \frac{1}{2}\frac{A^*}{m^*}\left(\frac{2}{3}L_w^*\phi_1\omega^*\right)^2 \frac{1}{N_w}\sum_{i}^{N_w}\left[v_w^{*2}\left(C_D \vec{d} + C_L \vec{\ell} \right) \right]^{(i)} - 1\vec{z} \tag{11}$$

Table 1: Dimensionless parameters

Parameter	Description	Range
$L_w^* = \frac{L_w}{L}$	Wing length ratio	[0.65, 0.85]
$A^*/m^* = \frac{N_w A_w \rho_{\text{air}} L}{m}$	Inverse wing loading	[0.2, 1.0]
$\omega^* = \omega\sqrt{\frac{L}{g}}$	Wing beat frequency	[8, 20]
γ_0	Wing stroke plane angle	[-180°, 180°]
ϕ_1	Wing flapping amplitude	[0°, 60°]
ψ_0	Mean wing pitch angle	[-60°, 60°]
ψ_1	Wing pitch amplitude	[0°, 60°]
δ_0	Wing pitch phase	[-180°, 180°]
σ_0	Fore/hind phase	[-180°, 180°]

3 Two Analytical Cases

3.1 Stationary Flight

3.1.1 Wingbeat-averaged Aerodynamic Force

In stationary flight, or *hovering*, the wingbeat-averaged aerodynamic force vector is opposed to and equal in magnitude to the weight vector. There are residual oscillations of the body due to fluctuations in the aerodynamic force over a wingbeat cycle, and inertial coupling between the body and wings (not modeled here). Assuming the corresponding body velocities are small compared to the characteristic wing velocity $\frac{2}{3}L_w^*\phi_1\omega^*$, we can approximate the mean aerodynamic force analytically by setting $\dot{x}^* = \dot{z}^* = 0$. As a result, the angle of attack is a simple function of the pitch angle ψ and the instantaneous sign of the flapping velocity:

$$\alpha = -\frac{\dot{\phi}}{|\dot{\phi}|}\left[\frac{\pi}{2} - \psi\right] \tag{12}$$

After making use of the appropriate trigonometric identities, this gives the lift and drag coefficients

$$C_L = -\frac{\dot{\phi}}{|\dot{\phi}|}C_{L,0}\sin 2\psi \tag{13}$$

$$\begin{aligned} C_D &= C_{D,0}\sin^2\psi + C_{D,\frac{\pi}{2}}\cos^2\psi \\ &= \frac{C_{D,\frac{\pi}{2}} + C_{D,0}}{2} + \frac{C_{D,\frac{\pi}{2}} - C_{D,0}}{2}\cos 2\psi \end{aligned} \tag{14}$$

The aerodynamic force normal to the stroke plane (along $\vec{n}$) is pure lift, and the force within the stroke plane (along $\vec{s}$) is pure drag. Denoting these two components as $F_{w,n}^*$ and $F_{w,s}^*$, respectively, they are

$$F_{w,n}^* = \frac{1}{2}\frac{A_w^*}{m^*}\left(\frac{2}{3}L_w^*\dot{\phi}\right)^2 C_L \tag{15}$$

$$F_{w,s}^* = -\frac{\dot{\phi}}{|\dot{\phi}|}\frac{1}{2}\frac{A_w^*}{m^*}\left(\frac{2}{3}L_w^*\dot{\phi}\right)^2 C_D \tag{16}$$

At this point, it is useful to linearize, in a sense, about the $\phi = 0$ position, by treating $\vec{s}$ as a fixed vector, parallel to the (x, z) plane. This is a common modeling choice, see *e.g.* [5]. The effect of the simplification is that force is increasingly overestimated as the wingbeat amplitude ϕ_1 increases. The effect is small for reasonable values of ϕ_1. Note the numerical model retains the moving $\vec{s}$ (wing beats along an arc, not a straight line) and we only perform the simplification for the analytical derivation, so that the time integral depends on ϕ_1 only as a simple scaling factor. Note further that the simplification has no effect on the mean if the wing pitch is symmetric for both halves of the stroke, since $\langle F_s^* \rangle$ would be zero regardless.

With the above simplification made, we introduce the the sweep kinematics $\phi(t^*) = \phi_1 \sin \omega^* t^*$, giving $\dot{\phi} = \phi_1 \omega^* \cos \omega^* t^*$. With the notation $\tau^* = \omega^* t^*$, we evaluate the wingbeat cycle integral $\langle \cdot \rangle = \frac{1}{2\pi} \int_0^{2\pi} (\cdot) d\tau^*$. An important question is how to deal with the sign flips due to the $\dot{\phi}/\phi$ factors appearing in both F_n^* (through C_L) and F_s^*. We can actually fold them into the "velocity squared" factors as follows

$$\begin{aligned} \frac{\dot{\phi}}{|\dot{\phi}|} \dot{\phi}^2 &= \dot{\phi} |\dot{\phi}| \\ &= \phi_1^2 \omega^{*2} \cos \tau^* |\cos \tau^*| \end{aligned} \tag{17}$$

It is also convenient to introduce the wingbeat amplitude shorthand $s_0^* = \frac{2}{3} L_w^* \phi_1$. Evaluating the integrals, and multiplying by the number of wings (A_w^* becomes A^*), we obtain the total cycle-averaged aerodynamic force components

$$\begin{aligned} \langle F_n^* \rangle &= -\frac{1}{2} \frac{A^*}{m^*} \left(s_0^* \omega^* \right)^2 \langle C_L |\cos \tau^*|^2 \rangle \\ &= -\frac{1}{2} \frac{A^*}{m^*} \left(s_0^* \omega^* \right)^2 C_{L,0} \langle \sin 2\psi \ \cos \tau^* |\cos \tau^*| \rangle \end{aligned} \tag{18}$$

$$\begin{aligned} \langle F_s^* \rangle &= -\frac{1}{2} \frac{A^*}{m^*} \left(s_0^* \omega^* \right)^2 \langle C_D \cos \tau^* |\cos \tau^*| \rangle \\ &= -\frac{1}{2} \frac{A^*}{m^*} \left(s_0^* \omega^* \right)^2 \frac{C_{D,\frac{\pi}{2}} - C_{D,0}}{2} \langle \cos 2\psi \ \cos \tau^* |\cos \tau^*| \rangle \end{aligned} \tag{19}$$

Note the constant drag term vanishes because $\langle \cos \tau^* |\cos \tau^*| \rangle = 0$. The expressions feature a common $\frac{1}{2} \frac{A^*}{m^*} \left(s_0^* \omega^* \right)^2$ factor, and a pitch- and aerodynamic-coefficient-dependent factor, which is lift-based for $\langle F_n^* \rangle$, and drag-based for $\langle F_s^* \rangle$. At this point, the only kinematics assumption we have made is sinusoidal flapping of the wings, while ψ remains generic. First, we can recover the analytical result of [5] for symmetric flapping with $\psi = -\frac{\pi}{4}$ for the $+\vec{s}$ stroke, and $\psi = +\frac{\pi}{4}$ for the $-\vec{s}$ stroke. Noting that $\cos \tau^* > 0$ for the $+\vec{s}$ stroke and < 0 for the $-\vec{s}$ stroke, and that $\langle \cos^2 \tau^* \rangle = \frac{1}{2}$, we recover the result

$$\langle F_n^* \rangle = \frac{1}{4} \frac{A^*}{m^*} \left(s_0^* \omega^* \right)^2 C_{L,0} \tag{20}$$

$$\langle F_s^* \rangle = 0 \tag{21}$$

We could obtain this result with our numerical model if we were to replace the sine wave with a square wave, and set $\psi_0 = 0$, $\psi_1 = \frac{\pi}{4}$. In reality, dragonfly wing pitch kinematics follow something between a sine wave and a square wave (*i.e.* a somewhat flattened sine wave). We stick to a sine wave here so as not to introduce higher harmonics ([6]) or a "flattening" coefficient ([7]), while keeping the more physical non-instantaneous stroke reversal (would be instantaneous with a square wave). To look at the effect of the pitch parameters on the force magnitude in the stroke plane and stroke-plane-normal directions, we introduce

$$q^* = \frac{1}{4} \frac{A^*}{m^*} \left(s_0^* \omega^* \right)^2 \tag{22}$$

$$C_{\psi,n} = -2 C_{L,0} \langle \sin 2\psi \ \cos \tau^* |\cos \tau^*| \rangle \tag{23}$$

$$C_{\psi,s} = -\left(C_{D,\frac{\pi}{2}} - C_{D,0} \right) \langle \cos 2\psi \ \cos \tau^* |\cos \tau^*| \rangle \tag{24}$$

such that $\langle F_n^* \rangle = C_{\psi,n} q^*$ and $\langle F_s^* \rangle = C_{\psi,s} q^*$. It is also convenient to introduce the overall magnitude $C_\psi = \sqrt{C_{\psi,n}^2 + C_{\psi,s}^2}$.

3.1.2 Power Efficiency

In addition to the mean aerodynamic force, another quantity of interest is the mean power developed over a wingbeat. Because we assume the wings are massless, power is only due to the work of the aerodynamic force. In the hovering case, only the drag force does work, because the lift force is strictly normal to $\vec{s}$. It follows the (dimensionless) wingbeat-averaged power is

$$\begin{aligned}\langle P^* \rangle &= \langle |F_s^* s_0^* \omega^* \cos \tau^*| \rangle \\ &= \frac{1}{2} \frac{A^*}{m^*} \left(s_0^* \omega^*\right)^3 \langle C_D |\cos \tau^*|^3 \rangle \end{aligned} \tag{25}$$

Following the derivation of the *cost of endurance* in [2], we can normalize the above by introducing the quantity $u_{\mathrm{ref}} = \sqrt{2mg/\rho A}$, more specifically its dimensionless form $u_{\mathrm{ref}}^* = \sqrt{2/\frac{A^*}{m^*}}$, giving

$$\begin{aligned}\frac{\langle P^* \rangle}{u_{\mathrm{ref}}^*} &= \frac{1}{2\sqrt{2}} \left[\frac{A^*}{m^*} \left(s_0^* \omega^*\right)^2\right]^{3/2} \langle C_D |\cos \tau^*|^3 \rangle \\ &= 2\sqrt{2} \left(q^*\right)^{3/2} \langle C_D |\cos \tau^*|^3 \rangle \end{aligned} \tag{26}$$

When hovering, the wingbeat-average net aerodynamic force $C_\psi q^* = 1 \implies q^* = 1/C_\psi$. We introduce the hovering power coefficient

$$C_{P^*} = 2\sqrt{2} \frac{\langle C_D |\cos \tau^*|^3 \rangle}{C_\psi^{3/2}} \tag{27}$$

which is analogous to the steady value $C_D/C_L^{3/2}$. The velocity-weighted drag coefficient in the numerator plays the role of C_D, and the force coefficient C_ψ in the numerator, which is, in general, part lift and part drag, plays the role of C_L. To be more specific, C_ψ is lift-only as long as the wing stroke is symmetric ($\psi_0 = 0$), which, for hovering, requires the stroke plane to be horizontal. Hovering at an inclined stroke plane requires, as we will see, $\psi_0 \neq 0$, and a drag contribution to C_ψ.

3.1.3 Force-Maximizing and Power-Minimizing Hover Kinematics

We show the dependence of C_ψ on the three parameters in Figure 3. Again, C_ψ here is obtained by taking $s_0^* = 0.25$, but this value has little effect on C_ψ as seen in the scatter plot from earlier. The result is what we expect, with ψ_1 slightly larger than 45° to provide a mean angle of attack that is closer to 45°. The dependence of F_a^* on δ_0 is symmetric but the force direction depends on the sign, we highlight $\delta_0 = +90°$ since with our conventions it results in a force in the direction of the leading edge. This optimal value of C_ψ is slightly smaller than $C_{L,0}$. Note that because the optimal pitch pattern is symmetric, the drag coefficient has no effect (drag component cancels out over the two halves of the wingbeat). We notice that with more wing elements, the optimal C_ψ is lower, converging to a value just above unity.

3.1.4 Effect of Morphology and Wingbeat Frequency

Now that we have identified two families of "optimal" wing pitch kinematics, independent of morphology and wingbeat frequency, we take a brief look at what our findings imply about these parameters. Using the hovering criterion $F^* = 1$, Figure 4 shows the required wingbeat amplitude s_0^* (function of L_w^* and ϕ_1) for a given $\left(\frac{A^*}{m^*}, \omega^*\right)$. The sweep amplitude ϕ_1 can be recovered by

$$\phi_1 = \frac{3}{2} \frac{s_0^*}{L_w^*}$$

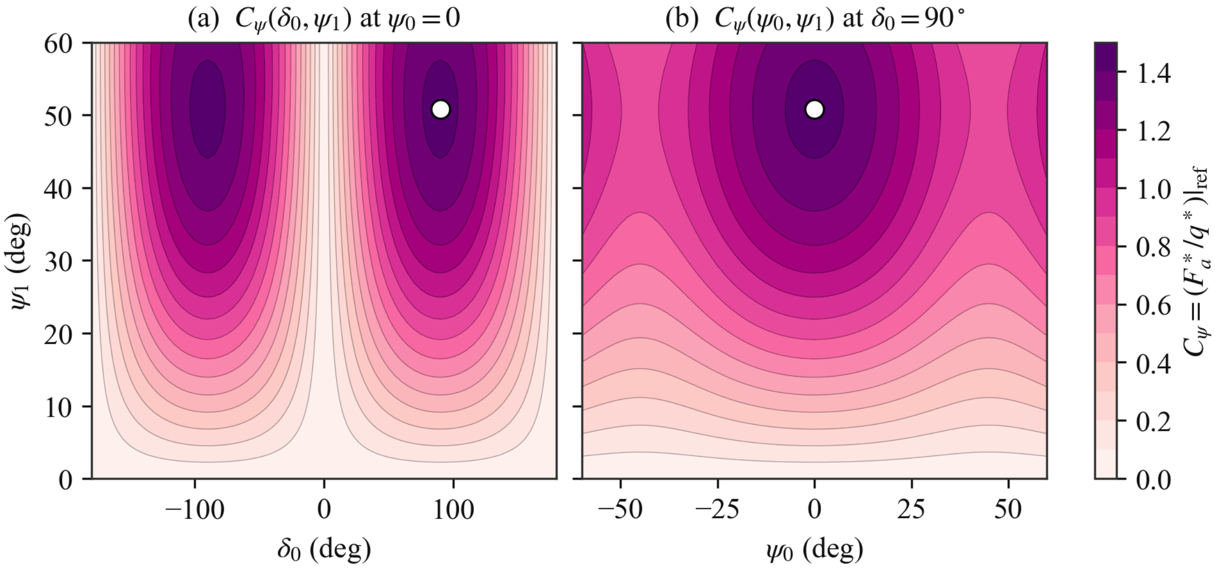

Figure 3: Dependence of C_ψ on pitch parameters at zero body velocity. Optimal value shown as white dots.

The very high amplitudes ($\phi_1 \approx 60^\circ$) associated with low values of both $\frac{A^*}{m^*}$ and ω^* suggest such combinations probably do not occur, and that the parameter space can be further restricted. The data source for ω^* does not report measurements necessary to obtain $\frac{A^*}{m^*}$, but we could maybe find other sources. We can estimate the typical hover wingbeat amplitude by assuming that the lowest ω^* occur at the highest $\frac{A^*}{m^*}$. For this set of parameters ($\omega^* = 8$ and $\frac{A^*}{m^*} = 1.0$) the amplitude is $s_0^* \approx 0.23$, which corresponds to $\phi_1 \approx 26^\circ$.

3.1.5 Hovering at an Inclined Stroke Plane

C_ψ may deviate from its "optimum" value. At this value, the wingbeat is symmetric, meaning the net force vector is normal to the stroke. This means hovering at the optimal C_ψ requires a horizontal stroke plane. Dragonflies however are known to hover with an inclined stroke plane, which could be due to efficiency considerations that we have ignored (picking the maximum C_ψ may not be exactly optimal energy-wise, even if it minimizes s_0, or our model simply does not capture other real effects that lead to the inclined stroke plane being advantageous). Without going too deep into these energy considerations at least for now, it is interesting at this point to consider the effect of the pitch parameters not just on the force magnitude but on its *direction* relative to the stroke plane. With an inclined stroke (angle γ_0 to the horizontal), hover (or vertical flight) requires the net aerodynamic force vector to be angled by $+\gamma_0$ to the stroke plane normal.

We denote the force direction angle relative to the stroke plane with β. We find numerically (and could probably show analytically) that β depends only on the mean pitch angle ψ_0, and on the sign of δ_0. To be exact, there is a slight variation in $\beta(\psi_0)$ over the range of ψ_1 and $|\delta_0|$ on the order of 0.1° to 1° depending on the wingbeat amplitude. We also find that the (ψ_1, δ_0) that maximizes C_ψ is invariant with ψ_0. We show $\beta(\psi_0)$ in Figure 5 along with the optimal C_ψ at a given ψ_0.

Three observations from the plot: (a) β is monotonic in ψ_0, so ψ_0 can be used conveniently as a control variable for the force direction, in addition to the stroke plane angle γ_0, (b) a 90° turn in ψ_0 reverses the force direction, so only moderate changes in ψ_0 are needed to achieve relatively large changes in β (there is factor of ~ 2), and (c) the wing stroke is most efficient when the mean pitch attitude points out-of-plane or in-plane, and least efficient when it forms a 45° angle with the stroke plane.

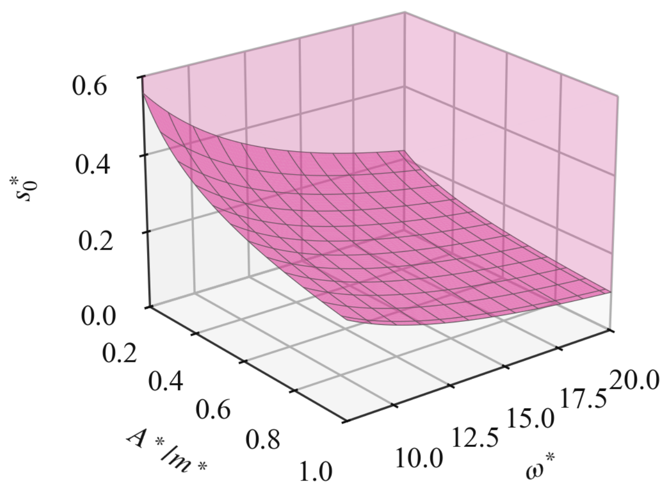

Figure 4: Approximate region of hover feasibility, with boundary surface.

3.1.6 Implications for Control

The above insights suggest a few possible ways to achieve hover control. Candidates for control variables can be split into two groups: those that control force magnitude (ϕ_1, ω^* through q^* and ψ_1, δ_0 through C_ψ), and those that control force direction (γ_0, ψ_0). Note this is not such a clean split, because ψ_0 also changes force magnitude through C_ψ, just more weakly than it changes direction. For simplicity, we would rather consider ω^* to be a fixed parameter, so we will exclude it at least for now. Since we also do not fully understand the factors that set the hovering stroke plane angle, we will also consider γ_0 to be fixed for now (later on, we compare the effectiveness of γ_0 and ψ_0 as directional control variables). In addition, it is beneficial to keep C_ψ at or near its maximum value, which sets ψ_1 and δ_0. This leaves ϕ_1 (or s_0^*) for magnitude and ψ_0 for direction. At equilibrium, ψ_0 will be such that $\beta = +\gamma_0$ (force is vertical), and ϕ_1 such that $F_a^* = 1$. Table 2 shows the required value of ψ_0 for different γ_0, for a reference configuration $L_w^* = 0.75$, $\frac{A^*}{m^*} = 0.6$, and $\omega^* = 14$. Different values of these parameters impose a different ϕ_1 for hover, which results in small changes in ψ_0 up to a few percent.

Table 2: Mean pitch for hover at various stroke plane angles for reference configuration

γ_0	ψ_0	$C_\psi^{\max}$
0°	0°	1.43
10°	−7.9°	1.40
20°	−15.1°	1.31
30°	−21.4°	1.21
40°	−26.7°	1.11

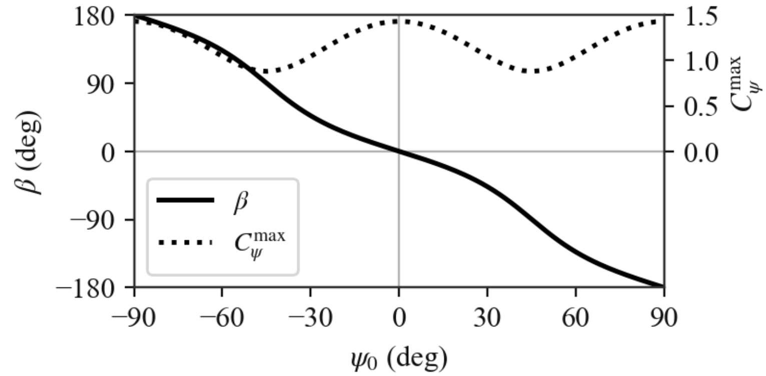

Figure 5: Effect of ψ_0 on force direction and wing stroke efficiency.

ϕ_1 is set by

$$\phi_1^{\text{hover}} = 3\left(L_w^*\omega^*\sqrt{C_\psi^{\max}(\psi_0)\frac{A^*}{m^*}}\right)^{-1}$$

The reference configuration used hereafter uses the following parameters (Table 3)

Table 3: Reference configuration

A^*/m^*	L_w^*	ω^*	γ_0
0.6	0.75	14	40°

3.2 Stroke-Plane-Normal Flight

3.2.1 Velocity Parameterization

We characterize the body velocity in the stroke-plane frame, by its angle χ from the stroke-plane normal and by the advance ratio

$$J = \frac{v_b^*}{s_0^*\omega^*}$$

i.e. the body speed measured against the wing speed scale. This choice has two motivations. First, the γ_0-invariance of the force magnitude carries over: $|\vec{F}_a^*|$ is unchanged when the stroke plane and the velocity vector are rotated together (verified numerically to machine precision), so a single map in (J, χ) covers all stroke plane angles. Second, scaling the speed with $s_0^*\omega^*$ keeps every velocity in the problem proportional to the wing speed, which preserves the q^* magnitude scaling: the effect of the body velocity can only enter through the coefficient, which becomes $C_\psi(\psi_0, \psi_1, \delta_0;\, J, \chi)$.

This last point assumes the separation $F_a^* = q^* C_\psi$ itself survives at nonzero velocity, which we check in Figure 6 by evaluating the scatter of F_a^*/q^* over random draws of $(\frac{A^*}{m^*}, \omega^*, \phi_1, L_w^*)$ at fixed pitch and fixed (J, χ). The residual is exactly zero for axial inflow ($\chi = 0°$ or $180°$) and grows with the *in-plane* velocity component, reaching about 7% at $J = 0.5$ and 11% at $J = 1$ for $\chi = 90°$. The distinction is geometric: an axial inflow meets the flapping wing at the same relative orientation at every point of the stroke arc, so the cycle geometry still depends on amplitude only through $s_0^*\omega^*$, while an in-plane wind makes an angle with the wing velocity that varies along the arc, re-introducing a weak dependence on ϕ_1 itself. The separation therefore remains useful well into the maneuvering regime, with the caveat that it is approximate for fast in-plane flight.

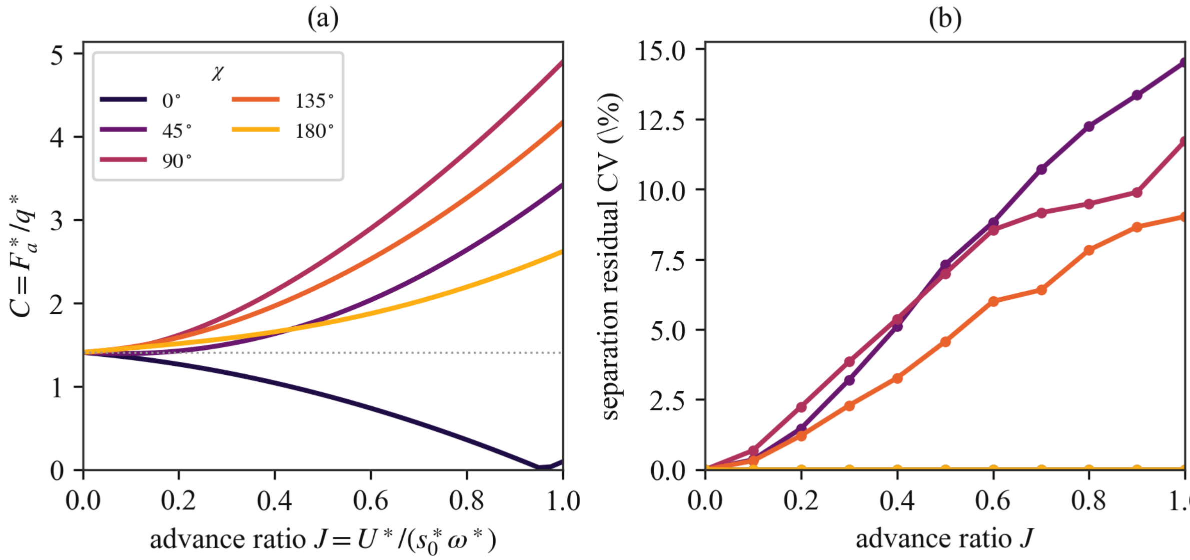

Figure 6: Survival of the $F_a^* = q^* C_\psi$ separation at nonzero body velocity. (a) C_ψ at nominal pitch as a function of advance ratio for several inflow directions χ. (b) Separation residual, evaluated as the coefficient of variation of F_a^*/q^* over random draws of the morphological and wingbeat parameters at fixed pitch.

3.2.2 Axial Inflow

The simplest slice of the velocity plane is axial inflow, $\chi = 0$, where the body moves along the stroke-plane normal. Dragonflies are observed to hold roughly this configuration, with the stroke-plane normal and the flight direction nearly aligned, so it is also the most relevant slice. It is the one case in which the zero-velocity analysis of **??** carries over essentially unchanged, because the body velocity has no component along the in-plane sweep direction $\vec{e}_\parallel$. The sweep term in v_s is then untouched by the body motion,

$$v_s = \tfrac{2}{3} L_w \dot{\phi}, \qquad v_n = \text{const}, \tag{28}$$

so the $\cos\phi$ coupling that produced the residual scatter at $\chi \neq 0$ disappears; this is exactly why the separation residual was found to be *zero* for axial inflow in Figure 6. The wing sees an oscillating sweep superposed on a uniform normal inflow—the classical heaving-plus-inflow kinematics of a blade element—and the angle of attack splits additively into the pitch and an inflow angle,

$$\alpha(\tau) = \psi(\tau) + \theta(\tau), \qquad \theta(\tau) = \operatorname{atan2}(v_s, v_n), \tag{29}$$

with $\tau = \omega t$. As the inflow vanishes ($J \to 0$) the inflow angle returns to $\theta \to \pm\frac{\pi}{2}$ and the hover result is recovered.

Writing the sweep-speed amplitude as $u = s_0\omega$ and the normal inflow through the advance ratio $J = v_b^*/(s_0^*\omega^*)$, we have $v_s = u\cos\tau$, $v_n = Ju$ for a body advancing along the cycle-mean force, and $v_w = u\sqrt{\cos^2\tau + J^2}$. Substituting into **??** and **??** and cycle-averaging in the small-ϕ limit—where, as in

??, the stroke basis is frozen and the average of the vector equals the vector of the averages—gives the moving-flight coefficient

$$C_\psi(\psi_0, \psi_1, \delta_0;\, J) = 2\sqrt{\langle \tilde{F}_s \rangle^2 + \langle \tilde{F}_n \rangle^2}, \tag{30}$$

$$\tilde{F}_n = \sqrt{\cos^2 \tau + J^2}\,\big(C_{L,0} \sin 2\alpha\, \cos \tau - J\, C_D\big), \tag{31}$$

$$\tilde{F}_s = -\sqrt{\cos^2 \tau + J^2}\,\big(C_D \cos \tau + J\, C_{L,0} \sin 2\alpha\big), \tag{32}$$

where $\tilde{F}_s, \tilde{F}_n$ are the stroke-frame components in units of $\frac{1}{2}\rho_{\text{air}} A_w u^2$ and α is given by Equation 29. The entire velocity dependence has collapsed onto the single parameter J: along the axial line the five-variable coefficient $C_\psi(\psi_0, \psi_1, \delta_0; J, \chi)$ reduces to a one-parameter family.

A parity argument sharpens the picture. For symmetric pitch ($\psi_0 = 0$, $\delta_0 = \frac{\pi}{2}$) the substitution $\tau \to \pi - \tau$ sends $v_s \to -v_s$, $\psi \to -\psi$, and hence $\alpha \to -\alpha$ at fixed v_n, so the $\tilde{F}_s$ integrand is odd and $\langle \tilde{F}_s \rangle = 0$: the cycle-mean force lies *exactly* along the stroke-plane normal at all advance ratios (verified numerically to $0.01°$). The aligned wingbeat therefore produces a force purely along the flight axis; redirecting it—for instance to support weight away from vertical flight—requires breaking the symmetry through the mean pitch ψ_0, the same lever used in hover. The coefficient then reduces to the single integral

$$C_\psi(J) = 2\left|\left\langle \sqrt{\cos^2 \tau + J^2}\,\big(C_{L,0} \sin 2\alpha\, \cos \tau - J\, C_D\big)\right\rangle\right|, \tag{33}$$

which returns to **??** as $J \to 0$ and decreases monotonically with J as the growing inflow detunes the angle of attack and bleeds off force—the wing losing authority as it advances into its own induced direction. The values track the numerical coefficient of Figure 11 along $\chi = 0$: $C_\psi = 1.43$ at hover, 1.21 at $J = 0.2$, and 0.90 at $J = 0.4$.

3.2.3 Power-Optimal Trim Across the Velocity Plane

For straight flight at body velocity $\vec{v}_b$ we solve for the power-optimal $\vec{u}$ producing weight-up, sweeping the redundant (γ_0, ψ_1) and trimming (ϕ_1, ψ_0) to the force at each node. The result (Figure 7) sets up the controller. The power-optimal stroke plane is *horizontal* ($\gamma_0 = 0$) for hover, forward flight and climb alike; only descent rewards a tilt. In forward flight ψ_0 does the steering, swinging from 0 to $-54°$ and saturating near its $-60°$ bound by $J \approx 0.9$; ψ_1 re-tunes *down* with inflow ($60°$ to $39°$), as the velocity maps anticipated; and ϕ_1 grows to hold the magnitude. The cost of forward flight rises steeply and monotonically (about ninefold by $J = 0.9$): with no induced-power reduction modeled, the drag dissipation simply grows with the cube of the relative speed, so there is no efficient cruise speed here. Vertical flight is gentler. Climb stays feasible by re-tuning ψ_1 (the infeasibility of the hover wingbeat at $\chi = 0$ is lifted once the pitch is allowed to adapt), at an aerodynamic cost that is mild until the climb work $\vec{v}_b \cdot \vec{F}_a^* = V_z W$ is added back to give the true metabolic cost; in descent the wing approaches an autorotative state, flapping little while the upward inflow carries the load.

4 Hover Dynamics and Control

4.1 Unsteady Dynamics

Before designing a controller, we look at how the actual, unsteady hover dynamics differ from the equilibrium picture we have been working with so far. Indeed the body velocity will not be exactly zero throughout, and will oscillate according to the fluctuations of the aerodynamic force over a wingbeat cycle. There is an additional effect due to small but nonzero mass of the wings, which will make the body bob up and down in the direction of the flapping motion. We have neglected this in our work so far. In this part we will evaluate the relative importance of the two effects on the body oscillations.

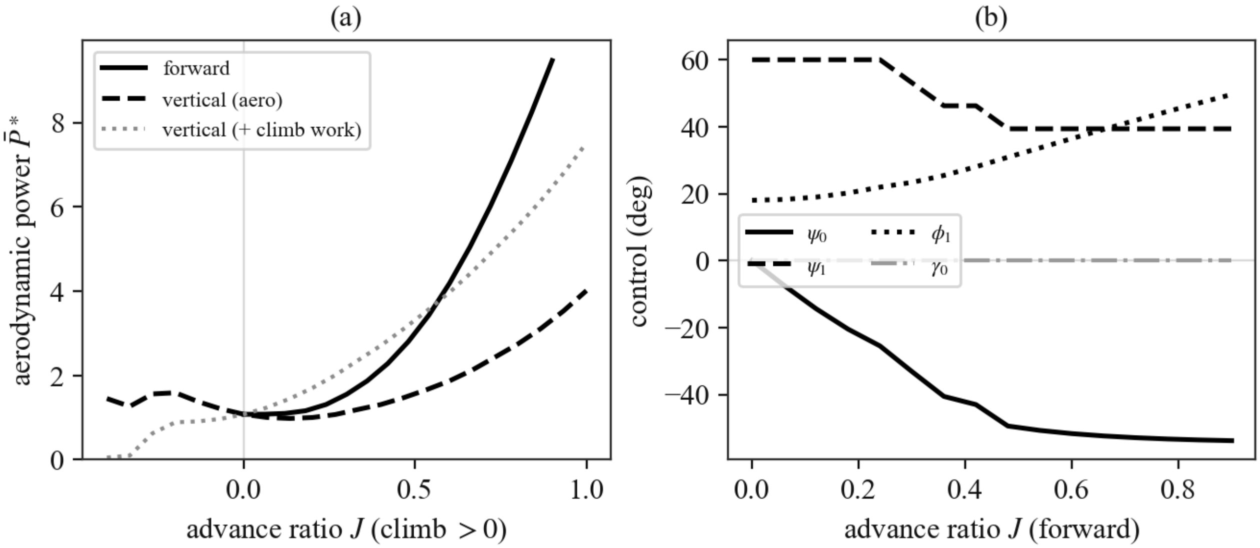

Figure 7: Power-optimal straight-flight trim vs. body velocity, reference configuration. (a) Aerodynamic power $\bar{P}^*$ vs. advance ratio for forward and vertical flight; the dotted curve adds the climb work $V_z W$ to the vertical aerodynamic power to give the metabolic cost. (b) The forward-flight control schedule: ψ_0 steers (and saturates), ψ_1 re-tunes down, ϕ_1 grows for magnitude, and the power-optimal γ_0 stays at zero.

A key parameter here is the fore/hind wing phase shift σ_0. We expect the oscillations to be more pronounced when the wings beat in-phase ($\sigma_0 = 0°$) than when they beat out-of-phase ($\sigma_0 = 180°$). In fact, if the forewing and hindwing motions are identical except for the phase shift, the wing inertia effect will be zero (true for translation, not for body pitch, which we do not consider in the present model).

We introduce a relative wing mass parameter $m_w^* = m_w/m$. According to [3], values range from 0.01 to 0.03 for *Anisoptera*. We run one hover wingbeat to obtain the body oscillation peak-to-peak amplitude, comparing the ideal massless wing case to the $m_w^* = 0.02$ case, for the reference configuration of Table 3 (Figure 8). The wing mass and aerodynamic contributions to the body oscillations appear to be of the same order, except, as expected, when the wings beat out-of-phase. We will consider $m_w^* = 0$ in the following, but should keep in mind that this is only effectively true at $\sigma_0 = 180°$.

Another effect of the aerodynamic force oscillations is positional drift. For a constant, open-loop wingbeat pattern, the dragonfly settles in a steady-state that has a small but nonzero mean body velocity, when the equilibrium prediction is that the zero mean force condition occurs at exactly zero mean body velocity.

Overall these effects are relatively mild, with instantaneous body velocities of approximately $0.25 s_0^* \omega^*$ in the worst case scenario ($\sigma_0 = 0°$, $\omega^* = 8$, $A^*/m^* = 1.0$), and only about $\frac{1}{10}$ of that for $\sigma_0 = 180°$, so the conclusions drawn from the equilibrium analysis should be applicable to controller design.

4.2 Hover Controller

Recall we identified ϕ_1 and ψ_0 as the two control variables for hover. ϕ_1 controls force magnitude, and ψ_0 controls force direction, with the secondary effect of reducing the force magnitude as it moves away from the stroke plane normal mean $\psi_0 = 0$. It seems natural to use ϕ_1 to control altitude, and ψ_0 to control fore-aft drift by tilting the force direction vector from the vertical. Specifically we propose the following control scheme

$$\psi_0 = \psi_0^{\text{eq}} - K_x(x - x_0)$$
$$\phi_1 = \phi_1^{\text{eq}} - K_z(z - z_0)$$

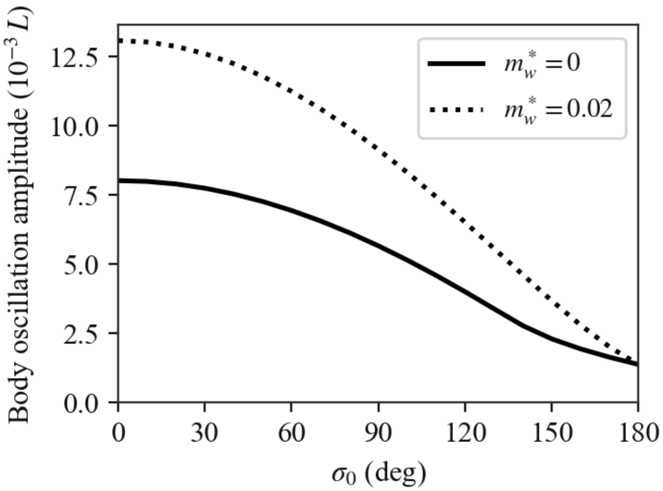

Figure 8: Effect of σ_0 and m_w^* on body oscillations during hover.

Note that orienting the force in the $-x$ direction requires pitching up (increasing ψ_0), so we will need $K_x < 0$. To design the proportional gains for the reference configuration (Table 3) we begin by forming the Jacobian matrix of the wingbeat-averaged aerodynamic force $\vec{F}_a^*$ at the hover equilibrium, using finite differences, to evaluate the control effectiveness of the two variables about the force axes. The Jacobian is

$$J = \begin{bmatrix} \dfrac{\partial F_{a,x}^*}{\partial \psi_0} & \dfrac{\partial F_{a,x}^*}{\partial \phi_1} \\ \dfrac{\partial F_{a,z}^*}{\partial \psi_0} & \dfrac{\partial F_{a,z}^*}{\partial \phi_1} \end{bmatrix} \approx \begin{bmatrix} -2.06 & -0.05 \\ +0.93 & 5.83 \end{bmatrix}$$

Notably ϕ_1 has a negligible effect on force tilt from the vertical ($F_{a,x}^*$), while as we pointed out before ψ_0 does have a significant secondary effect on $F_{a,z}^*$. Since ϕ_1 cleanly drives $F_{a,z}^*$, its control loop will absorb the ψ_0 cross term. The decoupled dynamics are (note mass matrix is I because of nondimensionalization)

$$\ddot{x}^* = -2.06 \times \delta\psi_0$$
$$\ddot{z}^* = 5.83 \times \delta\phi_1$$

So $\omega_n^2 = -2.06 K_x = 5.83 K_z$. We want ω_n slow enough that the controller sees the wingbeat-averaged force and not the fluctuations. Using $\omega_n = 1.5$ (10 times smaller than the reference wingbeat frequency $\omega^* = 14$) we get

$$K_x = -1.09$$
$$K_z = 0.39$$

The step response is shown in Figure 9. The figure shows the reference configuration along with low and high wing loading configuration. A^*/m^* was chosen as the variable here because it had the greatest effect on the damping of the response. The response is very noticeably more damped as A^*/m^* increases. Over the range of parameters, using the same gains designed for the reference configuration, the achieved ω_n was close to the design value of 1.5, and the response did not vary very much except along the A^*/m^* axis. The controller was stable across the full parameter space.

A more interesting result is the influence of stroke plane angle on damping, as shown in Figure 10. It appears that damping is strongest within the stroke plane, which means that vertical damping in the $\gamma_0 = 0°$

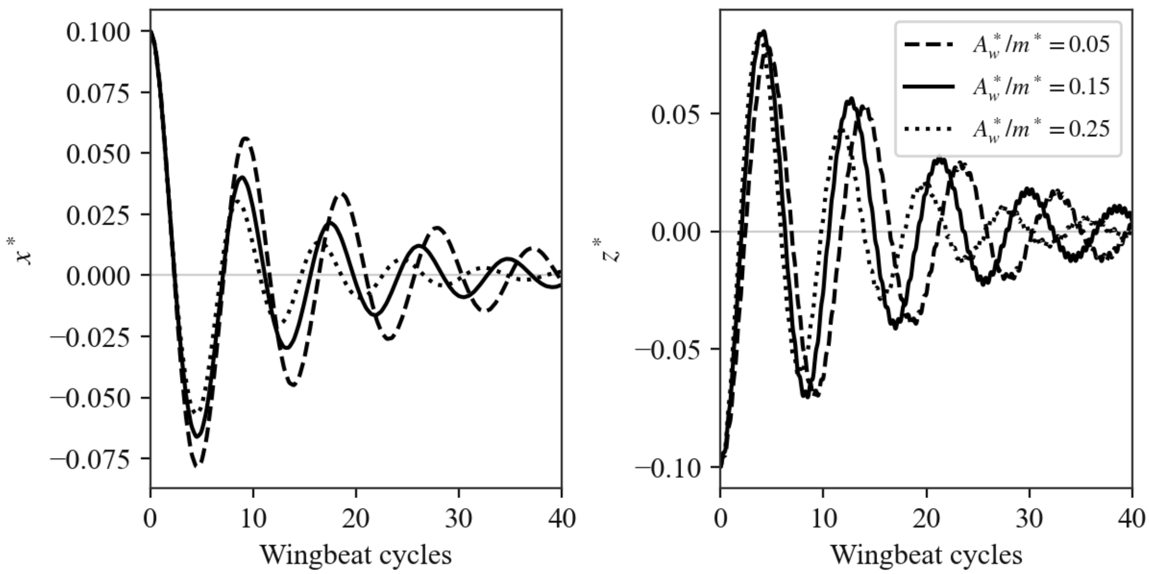

Figure 9: Effect of A^*/m^* on hover controller step response.

case is relatively low, and increases as the stroke plane is tilted. Horizontal damping decreases accordingly, with the end result being that there is a better match in the damping characteristics along both axes. Perhaps this is another advantage of hovering with an inclined stroke plane.

4.3 Robustness to Sensing Delay

We have assumed instant, perfect knowledge of body position. We now introduce a sensing delay and evaluate the constraints it imposes on controller design [needs to be written out].

5 Moving and Maneuvering Flight

So far we have focused on stationary flight, with at most small amplitude motions about a stationary or slowly drifting position. That is the case where the body velocity relative to the air is much less than the wing velocity relative to the body ($v_b^* \ll s_0^*\omega^*$). Since we are interested not just in hover but in maneuvering flight, we need to consider the case where $v_b^* \sim s_0^*\omega^*$, perhaps even $v_b^* \gg s_0^*$.

5.1 C_ψ over the Velocity Plane

With (J, χ) added to the three pitch parameters, C_ψ is a function of five variables, which we cannot map directly. Three structural observations (all numerical) reduce the visualization problem to a manageable size:

1. Mirror symmetry: $C_\psi(J, \chi, \psi_0) = C_\psi(J, -\chi, -\psi_0)$ to machine precision, so at $\psi_0 = 0$ the velocity plane is symmetric about the stroke-plane normal and only a half-plane needs to be computed.

2. The optimal pitch phase stays pinned at $\delta_0 = 90°$ across the entire velocity plane.

3. Consequently only ψ_1 retunes with velocity, and it does so smoothly.

Since (J, χ) is just the body velocity vector expressed in the stroke-plane frame, we can treat the velocity plane as a disk (radius J, angle χ) and show each scalar of interest as one polar map (Figure 11): (a) C_ψ at the

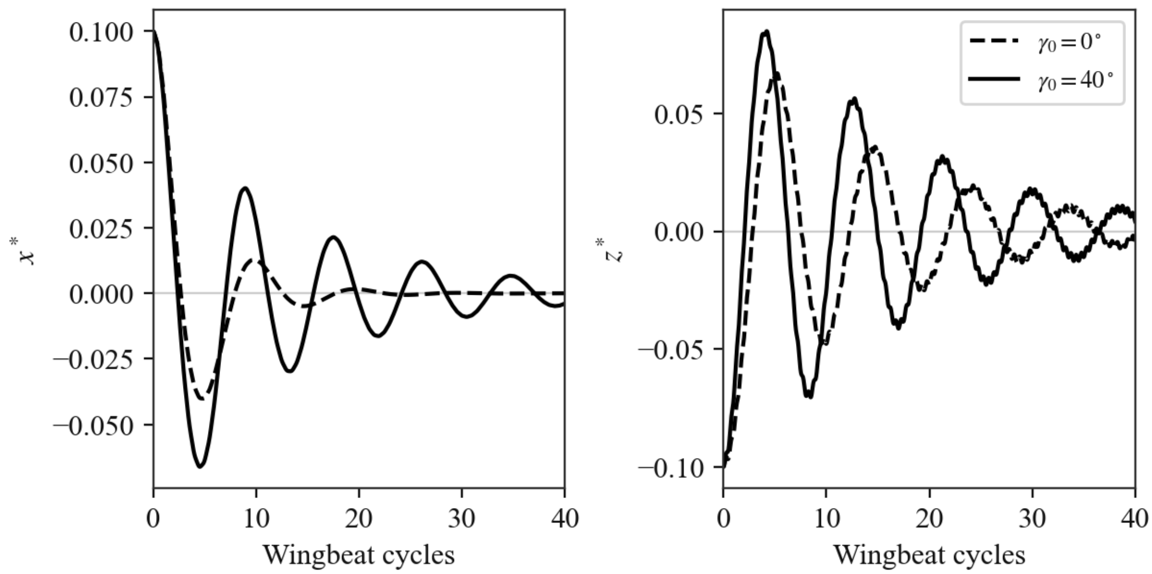

Figure 10: Effect of γ_0 on hover controller step response.

hover-optimal pitch, i.e. how the hover wingbeat performs when the body moves; (b) C_ψ with ψ_1 re-optimized at each velocity within its allowed range; and (c) the re-optimized ψ_1 itself.

The maps show three regimes:

- **Axial inflow** (**χ near** $0°$): the inflow through the stroke plane washes out the angle of attack and the hover wingbeat loses its force rapidly, from $C_\psi = 1.43$ at hover to 0.90 at $J = 0.4$ and essentially zero (0.04) by $J = 0.8$. Re-tuning recovers force (1.17 at $J = 0.4$, 1.00 at $J = 0.8$) by *reducing* ψ_1 to keep the section near its best angle of attack in the combined flow.

- **In-plane wind** (**χ near** $\pm 90°$): the mean-square relative velocity grows as the wind adds to the wing speed on one half-stroke faster than it subtracts on the other, and C_ψ *increases* strongly (2.10 at $J = 0.4$, 3.66 at $J = 0.8$ at hover pitch). The hover wingbeat is already close to optimal here; re-tuning adds only a few percent.

- **Reverse axial inflow** (**χ near** $180°$): the inflow steepens the angle of attack, and the optimum saturates at the upper bound of the pitch amplitude range ($\psi_1 = 60°$).

Between the in-plane and reverse-axial regimes, around $\chi \approx \pm 110°$ at high J, the optimal ψ_1 switches branches abruptly (from near-zero pitch amplitude, where the wing behaves like a fixed-pitch blade in a dominant wind, to the large-amplitude branch), visible as the sharp boundary in panel (c).

For equilibrium, since the model has no body drag, steady flight at any velocity requires the same force as hover: $F_a^* = 1$ directed vertically. The magnitude condition is $q^* C_\psi = 1$, so panel (a) divides the velocity plane into regions where the hover wingbeat produces a force surplus that maneuvering can spend (C_ψ above its hover value, roughly the whole lower half-plane) and regions where flight demands either a larger q^* (faster or wider flapping) or a re-tuned ψ_1 (the dashed $C_\psi = 1$ lobe around $\chi = 0$ marks where climbing flight with the reference $q^* = 1$ becomes infeasible at hover pitch). The direction condition is taken up next: at nonzero velocity the force direction β is no longer a function of ψ_0 alone but acquires a strong (J, χ) dependence (e.g. at $\chi = 90°$, $J = 0.3$ the force tilts by $-31°$ at $\psi_0 = 0$), which is the central complication for the maneuvering controller.

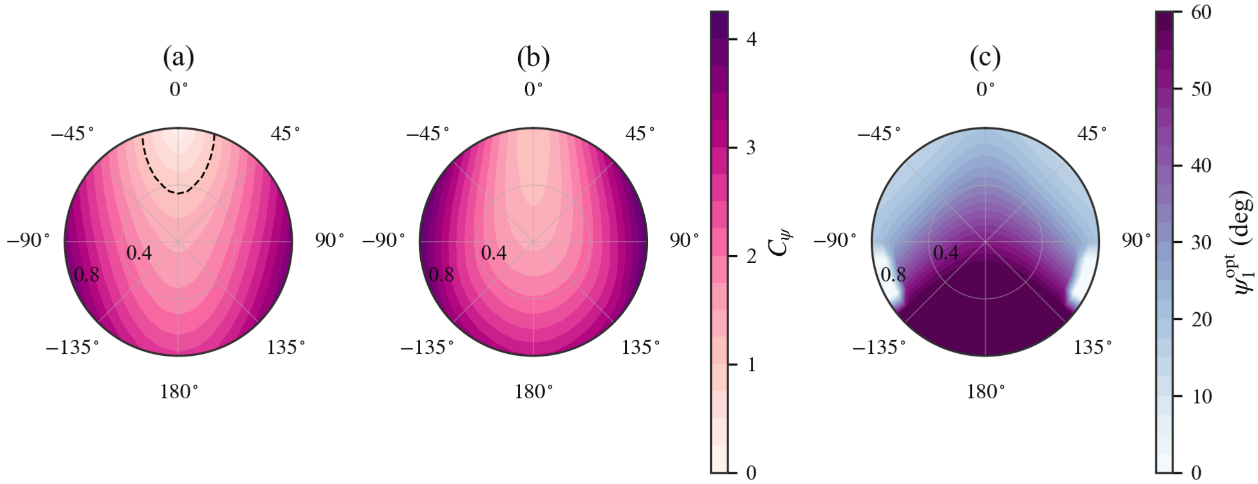

Figure 11: C_ψ over the body-velocity plane (radius: advance ratio J, angle: χ from the stroke-plane normal), at $\psi_0 = 0$, $\delta_0 = 90°$, for the reference configuration. (a) Hover-optimal pitch ($\psi_1 = 51°$), dashed line marks $C_\psi = 1$. (b) ψ_1 re-optimized at each velocity over the range of Table 1. (c) The re-optimized ψ_1.

5.2 Control for Maneuvering Flight

Two features of the model shape the whole control problem. First, with no body drag, steady straight flight at *any* velocity needs the same net force as hover—the weight, pointing up—so there is no trim curve to chase with speed; "cruise" is just a re-trim of the controls as the wing comes to see inflow, and maneuvering proper (accelerating, changing flight path) is a transient excursion of the force vector away from weight-up. Second, the cycle-averaged force is a velocity-scheduled map $\vec{F}_a^*(\vec{u}; \vec{v}_b)$ of the controls $\vec{u}$, of which the hover controller of the previous section is the $\vec{v}_b = 0$ slice. The maneuvering controller is therefore the hover controller scheduled on velocity, and the new content lives entirely in the velocity dependence already mapped in Figure 11.

We keep $\delta_0 = 90°$ (optimal everywhere) and use four controls: the two hover handles ϕ_1 (magnitude) and ψ_0 (direction), plus the two redundant degrees of freedom ψ_1 and γ_0. We resolve the redundancy by minimizing the cycle-averaged aerodynamic power—the operative form of the cost of endurance P^*, which it equals up to the constant weight-support normalization.

5.3 Directional Authority: ψ_0 vs. γ_0

The previous section flagged that at velocity the force direction is no longer set by ψ_0 alone. We can now compare the two directional controls quantitatively. At hover ψ_0 is the better lever, tilting the force by 1.25° per degree against 1.00° per degree for γ_0, and it has its full range available. As forward speed grows, the two trade places: ψ_0's authority falls (to 0.80° per degree by $J = 0.8$) while γ_0's rises (to 1.26°), crossing near $J \approx 0.5$. Worse, the ψ_0 *trim* itself marches toward its bound (to −55°, leaving 5° of headroom by $J = 0.8$), so it runs out of steering range just as it loses sensitivity. This is why we hold γ_0 horizontal at low speed, where ψ_0 is both more effective and unsaturated, and reserve γ_0 as the directional control for fast forward flight, where ψ_0 can no longer do the job.

5.4 Closed-Loop Maneuvers

The controller is a first-order velocity tracker with feedforward, $\vec{a}_{\text{des}} = \vec{a}_{\text{ref}} + K_v(\vec{v}_{\text{ref}} - \vec{v}_b)$, whose required force $\vec{F}_{\text{des}} = \vec{a}_{\text{des}} - \vec{g}$ is realized by dynamic inversion of the cycle-averaged map: hold $\gamma_0 = 0$, re-optimize ψ_1 for power, and trim (ϕ_1, ψ_0) to $\vec{F}_{\text{des}}$ at the current velocity. As at hover, the control is updated once per wingbeat—an order of magnitude below ω^*—held across the beat, and the *unsteady* instantaneous force is integrated, so the run is a genuine test of the cycle-averaged design.

Figure 12 shows two maneuvers. In the forward dash the body tracks an accelerate–cruise–decelerate velocity command to $J \approx 0.5$ and back, with only a small within-cycle ripple at cruise (standard deviation ≈ 0.018 in v_x^*) and a peak tracking error of 0.19 confined to the hardest acceleration phase. In the pull-up the velocity command rotates from forward to vertical, and the body follows it through a smooth quarter-turn in the sagittal plane, with ψ_0 steering from $-42°$ to near 0 as the flight path swings into the climb (peak speed error 0.04). Both confirm that the velocity-scheduled, power-optimal inversion controls maneuvering flight across the regimes where the inflow strongly reshapes the force map.

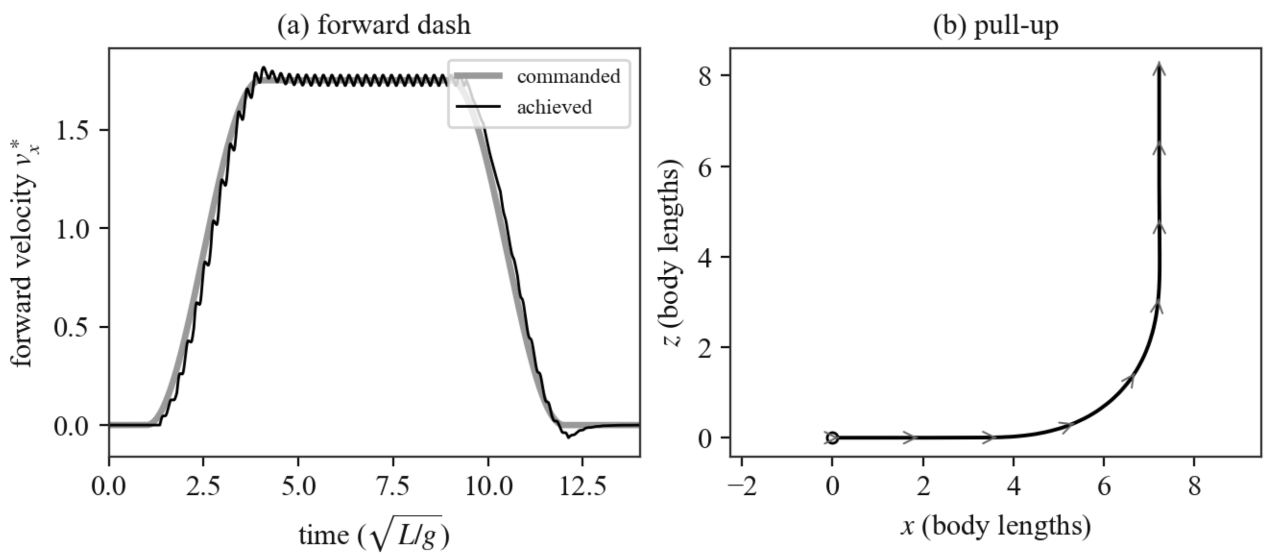

Figure 12: Closed-loop maneuvers (unsteady simulation, control updated once per wingbeat). (a) Forward dash: commanded vs. achieved forward velocity through accelerate–cruise–decelerate. (b) Pull-up: the sagittal-plane path, with body-velocity arrows rotating from horizontal to vertical.

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